The Stack

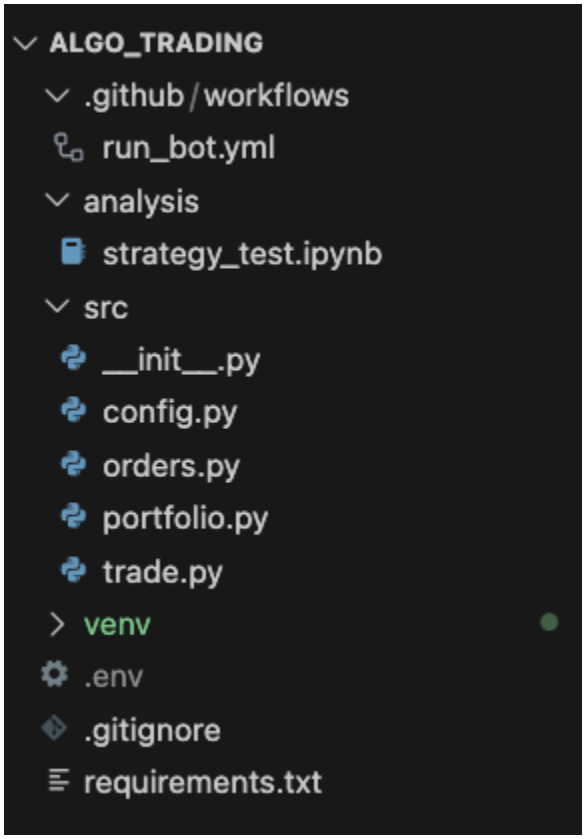
- Alpaca: The broker API.
- GitHub: Hosts the code and runs the daily automation.
- VS Code: Where I wrote the Python (you can use Google Colab, PyCharm, etc.).

Step 1: Alpaca Setup

1. Go to Alpaca Markets (sign in) and toggle "Paper Trading" on the dashboard.
2. Generate your API Keys on the right side.
3. Save the Key ID and Secret Key immediately. You need these to connect your code to the account.
    - Note: You can still buy/sell manually on the Alpaca dashboard if you ever want to override the bot.

Step 2: The Code Structure

I organized my folder (algo_trading) to keep things modular rather than a single, messy script.



- src/config.py: Handles the login. It connects to Alpaca using the keys so I don't have to rewrite the connection logic in every file.
- src/trade.py: The main logic. It checks my current portfolio first to see if I already hold the position. If not, it places a buy order.
- src/orders.py: A helper file where I stored functions like buy_market so the main strategy file stays clean.
- .env: This is where I pasted my API keys locally so they aren't hardcoded in the script.

These are the names I gave my folders, by the way. Note that I have code in all of these files. I am trying to create a system for buying/selling, so I have to write minimal code when writing an algorithm or

analyzing an asset. There's a cookie-cutter data science structure I have used before, but I think it's too complicated for what I am doing right now.

Step 3: Security

Never upload API keys to GitHub.

- Locally: I used a .env file and added it to .gitignore.
- On GitHub: I went to Settings > Secrets and variables > Actions and added my keys (ALPACA_KEY, ALPACA_SECRET) there. This lets the bot access them safely without hackers scraping them from the code.

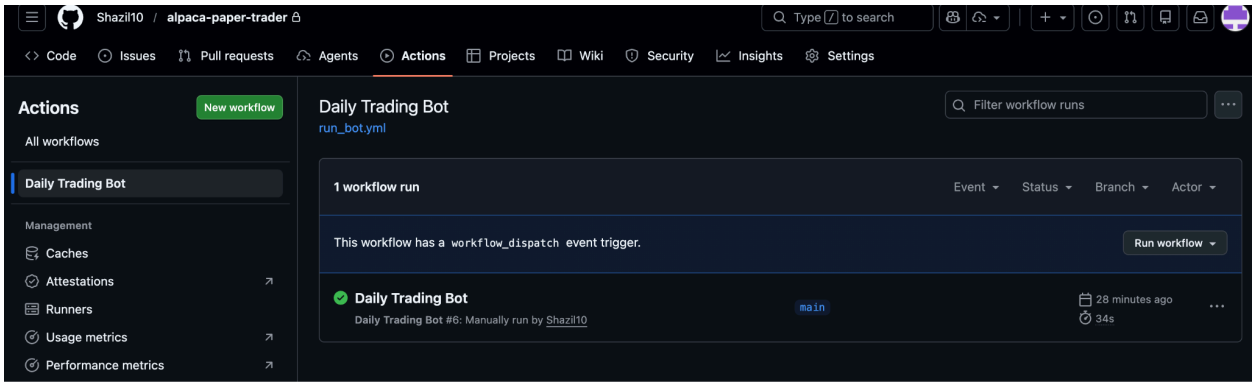
(I know that it's just paper trading, but I thought it's a good thing to do in case I might use similar code / do similar steps to go live trading)

Step 4: Automating it with GitHub Actions

I didn't want to run the script manually every day, so I used GitHub Actions. I created a YAML file in .github/workflows/ that tells GitHub to:

1. Wake up every weekday morning
2. Install Python and my dependencies.
3. Run python src/trade.py.

(930 AM EST to 4 PM EST, i..e the market timings) Now it runs entirely in the background.



A Note on Visuals

Since this runs in the terminal/cloud, there are no charts. If you want to see pretty graphs of your performance, you can just connect your Alpaca account to TradingView to visualize the data while the Python script handles the execution. Of course, you can write Python code on .ipynb and make visualizations of your own portfolio that way! I'll do that too.